Missing data is one of the most common challenges in datasets. There are various ways to handle missing values, depending on the nature of the dataset and the goals of the analysis.

Removing Rows or Columns with Missing Values

In cases where missing values are minimal and non-critical, you can remove the rows or columns containing them: import pandas as pd

data.dropna(inplace=True)

While this method is straightforward, it is not always ideal, especially if the removed data represents a significant portion of the dataset.

Imputing Missing Values

Imputation involves filling in missing values with a substitute value, such as the mean, median, or mode of the column: data['column_name'].fillna(data['column_name'].mean(), inplace=True) For categorical data, you might use the mode:

data['column_name'].fillna(data['column_name'].mode()[0], inplace=True) Advanced imputation methods include using algorithms like K-Nearest Neighbors (KNN) to predict missing values based on other features.

Removing Duplicates

Duplicate records can distort analysis results, particularly in aggregations or statistical computations.